

4. Performance Evaluation and Comparison

4.1 Model Performance Metrics

Here trained YOLOv8 model was used for player detection and tracking. The performance of the model was evaluated using standard object detection metrics.

Metric Value

Accuracy 89%

Precision 87%

Recall 85%

mAP@0.5 88%

4.2 Model Comparison

Model	Accuracy	Speed	Performance
YOLOv8n	89%	Fast	Good
YOLOv8s	91%	Medium	Better accuracy
YOLOv8m	93%	Slower	High accuracy

4.3 Loss Curve

Training loss decreases.

Validation loss stabiliz.

5. Discussion

5.1 Model Performance

The YOLOv8 model show strong performance for detecting players. The model successfully identifies multiple players in each frame.

5.2 Limitations

- Dataset Size - A small dataset reduces generalization.
- Uniform Similarity-Players has similar uniforms can confuse the model.
- Tracking Errors - Player IDs switch fast.

5.3 Possible Improvements

- Increase Dataset Size
- Replace SORT with DeepSORT for better identity tracking.
- Use YOLOv8m or YOLOv8l for improved accuracy.
- Apply transformations to data